Prabir Kumar Das

Biomedical Engineer

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"Translating individual-level brain modelling into personalised and sustainable brain health solutions."

Marie Skłodowska-Curie (MSCA) Doctoral Researcher in the **EU BRAINET Doctoral Network** at Tampere University, Finland . Biomedical engineer with strong expertise in AI-based medical imaging, including diffusion models, functional connectivity analysis, and inverse problem solving. Develops physics-informed and interpretable AI methods for ultrasound, fMRI, and EEG, with a focus on biomarker discovery, disease monitoring, and robust clinical decision support for neurological disorders.

Education

2026 – 2029 **Marie Skłodowska-Curie Doctoral Researcher (PhD), Biomedical Technology**, Faculty of Medicine and Health Technology, Tampere University, Finland

- **MSCA Fellow**, **EU BRAINET Doctoral Network** (Marie Skłodowska-Curie Actions, Horizon Europe) — a competitively selected European doctoral network. Supervisor: Prof. Jari Hyttinen.
- Developing electrophysiological methods (MEA, EEG) to assess the effects of non-invasive temporal-interference (TI) stimulation on the nervous system *in vitro* and *in vivo*, including control of stimulus patterns from novel networked microdevices.
- **Planned secondments**: TU Ilmenau and NEUROCONN, Germany (3 months).

2024 – 2025 **Master of Science and Technology in Complex System Engineering (Biomedical Track)**, GPA: 3.55/4, Centrale Méditerranée, France

2020 – 2022 **M.Tech in Medical Imaging and Informatics**, CGPA: 8.86/10, SMST, Indian Institute of Technology Kharagpur, India **(QS rank: 215)**

2012 – 2016 **B.Tech in Electronics and Instrumentation**, CGPA: 8.29/10, West Bengal University of Technology (MAKAUT), Kolkata, India

Research & Project Experience

Neuroscience, EEG & Quantitative Modeling

2021 – 2022 **Master's Thesis Student** — IIT Kharagpur, Medical Informatics Lab (**Supervisor**: Dr. Debashree Guha Adhya)

Thesis: Functional Brain Connectivity Analysis with Brain–Behavior Correlation: An EEG-Based Study — Analyzed task-based and resting-state EEG using graph-theoretic and phase-synchronization models to study small-world topology, hub formation, and brain–behavior correlations, establishing EEG network biomarkers relevant for longitudinal neurophysiological monitoring. (GitHub Repo) (YouTube)

07/21 – 10/21 **Summer Research Internship** — IEEE EMBS Society, IIT Kharagpur (Certificate)

Project: fMRI Visual Eccentricity Analysis — MATLAB-based analysis of BOLD signals to quantify variability and mean activation across visual-field eccentricity.

Sep 2024 – Feb 2025 **Team Project** — supervised by **Serge Mesure** (Aix-Marseille Université) & **Didier Bertrand** (CryoKiné), Institut des Sciences du Mouvement, Marseille

Project: Design of Biomedical Instrumentation for Human Movement Analysis. (Report)

Deep Learning in Medical Imaging

Mar 2025 – Aug 2025 **Master's Thesis Internship** — MINDS (coMputational imagINg anD viSion) Research Team, IRIT, Toulouse (**Supervisors**: Dr. Duong-Hung Pham, Dr. Denis Kouamé) (Website)

Thesis: Ultrasound Image Deconvolution via Block-Circulant Spectral Decomposition in Diffusion Models — Developed a forward-model-based inverse-problem framework using PSF/BCCB operators and SVD-domain sampling (DDRM) for robust image restoration under realistic degradations. (GitHub)

2025 – Ongoing **Research Collaborator, Deep Learning in Medical Imaging** — with Prof. Swati Rani Hait, Department of Mathematics, BITS Pilani

Project: Adaptive Genetic Algorithm–Based Deep Feature Selector for Cancer Detection in Lung Histopathological Images — Hybrid framework combining channel-attention CNN features with an adaptive Genetic Algorithm for feature selection, followed by KNN classification; achieved 99.75% accuracy on the LC25000 dataset. (GitHub)

Publications & Intellectual Property

Peer-Reviewed Conference Papers

PAUL 2026 — **Das P K**, Shaikh M U A B, Hait S R, Bora D J, Nath D. "*Multi-Domain EEG Characterization of Anxiety Severity in the DASPS Database: Robust Alpha Markers, Situation-Resolved Engagement Dynamics, and the Limits of Participant-Level Classification.*" 9th International Conference on Pattern Analysis, Understanding and Learning (PAUL 2026), Springer CCIS Conference Proceedings. [**Accepted**]

PAUL 2026 — **Das P K** (Corr.), James R, Kour H, Nath D, Bora D J. "*Beyond Point Estimates: A Bootstrap-Validated Benchmark for Class-Imbalanced Cell Detection in Vaginal Cytology.*" 9th International Conference on Pattern Analysis, Understanding and Learning (PAUL 2026), Springer CCIS Conference Proceedings. [**Accepted**]

IEEE CVMI 2026 — **Das P K** et al. "*Advanced EEG-Based Schizophrenia Detection Using CNN+BiLSTM with Novel Biomarkers and Explainable AI.*" 5th IEEE Int. Conf. on Computer Vision and Machine Intelligence, COEP Technological University, Pune. [**Published**]

IC-SIT 2026 — Pandey S, **Das P K** (Corr.), Sharma R. "*EEG Correlates of Music Familiarity and Enjoyment: Individual-Level Analysis.*" 2nd Int. Conf. on Intelligent Computing and Sustainable Innovation in Technology (IC-SIT). [**Corresponding Author —Presented**]

INCOWOCO 2026 — Salunke A S, Shetty K, Kour H, Jalan P, **Das P K** (Corr.). "*Uncertainty-Aware Benchmarking, Attribution Agreement, and a Pre-Ultrasound Screening Model for Polycystic Ovary Syndrome Prediction.*" 3rd IEEE International Conference for Women in Computing (INCOWOCO 2026), Symbiosis Institute of Computer Studies and Research, Pune, India. [**Accepted**]

AICTA 2026 — Wayal A, **Das P K**, Sahoo S, Nath D. "*Deep Learning-Based Anterior and Posterior Hippocampus Segmentation in Brain MRI: A 3D Residual Ensemble Approach.*" 4th Int. Conf. on Artificial Intelligence, Computing Technologies and IoT, NIT Silchar (Springer). [**Presented**]

IEEE TEMSMET 2026 — **Das P K** et al. "*Quality-Aware Fetal Ultrasound Plane Classification with Uncertainty Quantification and Calibration.*" IEEE Int. Conf. (TEMSMET), Hyderabad, India. [**Accepted**]

IEEE TEMSMET 2026 — Wayal A, Nath D, **Das P K** (Corr.) et al. "*Deep Learning-Based Peripheral and Transition Zone Segmentation in Prostate MRI: A 3D Ensemble Approach.*" IEEE Int. Conf. (TEMSMET), Hyderabad, India. [**Corresponding Author — Accepted**]

IEEE INSTCon 2026 — Byahatti H S, **Das P K**, Nath D, Hait S R. "*A Comparison of CNN Architectures for IDC Identification Using Transfer Learning on Enhanced VGG19.*" 1st IEEE Int. Conf. on Instrumentation, NIT Rourkela. [**Presented**]

IEEE INSTCon 2026 — Parkhedkar M, **Das P K**, Nath D, Hait S R. "*Colon Cancer Segmentation Using Deep Learning: A 3-D nnU-Net Pipeline with Cross-Validation.*" 1st IEEE Int. Conf. on Instrumentation, NIT Rourkela. [**Presented**]

SIPCOV 2026 — **Das P K**, Tripathi S, Sahoo S, Mukherjee K, Nath D. "*Mapping the Visual Periphery: Eccentricity-Dependent Patterns in Human Visual Cortex.*" 3rd Int. Conf. on Signal Processing and Computer Vision (SIPCOV), Oral Presentation. [**Presented**]

SIPCOV 2026 — **Das P K**, Darekar S (Corr.), Ramchandani K G, Deshpande N M, Nath D, Mukherjee K. "*Liver Tumour Segmentation Using Deep Learning: A 2.5D U-Net Baseline and nnU-Net 2D Ensemble.*" 3rd Int. Conf. on Signal Processing and Computer Vision (SIPCOV), Oral Presentation. [**Presented**]

Registered Designs (Intellectual Property)

2026 — **Registered Design** — "*Pediatric Skin Cancer Diagnostic AI Device.*" Design No. 493775-001, Class 24-01; registered 20 May 2026. Co-inventor (2nd of 10 applicants).

Technical Skills

Signal Processing	EEG preprocessing and artefact removal; ECG analysis; spectral analysis (PSD, band power); time–frequency methods; phase synchronisation (PLV); functional connectivity modelling
Computational Neuroimaging	EEG/fMRI connectivity analysis; graph-theoretic modelling (small-worldness, efficiency, modularity); brain–behaviour modelling; LOSO cross-validation
AI & ML	Deep learning (CNNs, ResNet, U-Net); diffusion models (DDPM, DDRM); classical ML (SVM, Random Forest, XGBoost); interpretable AI (SHAP, Grad-CAM); biomarker discovery and validation
Computational Imaging	Inverse problems; physics-informed modelling; PSF-based forward models; spectral methods (FFT, SVD); medical image reconstruction; MRI (qMRI, DICOM)
Programming & Computing	Python (PyTorch, NumPy, SciPy), MATLAB, C/C++; GPU computing (CUDA); OpenCV, NiBabel, NetworkX; Linux, Git, LaTeX

Work Experiences

Oct 2025 – Aug 2025 **Assistant Professor**, *D.Y. Patil International University*, Pune, India, Department of Bioscience and Bioengineering
Teaching: Advanced Computing for Biological Systems; Biomedical Signal Processing; Medical Imaging.
Supervision: Mentoring undergraduate research projects.
Administrative Roles: Internship coordination; Skill Development Committee (In-Charge); Organizing Committee member for departmental conference and Biospark Hackathon; Organized industrial (Cytiva) and hospital visits.

Mar 2023 – Jul 2023 **Senior Mathematics Faculty**, *ORCHIDS The International School, India*
Taught advanced mathematics to high-school students; assessments, lesson planning, parent–teacher coordination. (Website) (Training Certificate)

Aug 2021 – Dec 2021 **Teaching Assistant**, *IIT Kharagpur — MATLAB for Quantitative Techniques in Medicine (MM61311)*
Conducted MATLAB labs on medical imaging and quantitative modeling. (GitHub)

Nov 2016 – Aug 2018 **Assistant Systems Engineer**, *Tata Consultancy Services (TCS), India*
Software development and automation for enterprise clients and cross-team collaborations. (Service Certificate)

Awards & Achievements

- **Marie Skłodowska-Curie Actions (MSCA) Doctoral Fellowship**, Horizon Europe — selected as a Doctoral Candidate in the **BRAINET Doctoral Network**, hosted by Tampere University, Finland (2026–2029).
- **TIGER Master's Excellence Scholarship** of **€10,000** for the academic year 2024–2025 (Award Letter).
- Secured **All India Rank 140** in GATE 2020 **(Graduate Aptitude Test in Engineering)** from Instrumentation Engineering (IN) with **98.09 percentile** (Certificate).
- Received **Post Graduate Scholarship** of **INR 2,97,600** from the All India Council for Technical Education under the Department of Higher Education, **Government of India**.

Summer Schools and Industrial Training

11/24-12/25 **MULTIWAVE MRI Company** on "SILENCE: Acoustic Metamaterial for MRI Noise Cancellation "

04/21-05/21 Accelerated Data Science by **NVIDIA** (IIT Kharagpur micro-credit course)

03/2021 **Advances in Medical Imaging** by Indian Institute of Technology (BHU)(Certificate)

JULY 2022 **Neuromatch Academy**, (online) **Computational Neuroscience** (Certificate)

Project: Analysing emotion processing in brain using fMRI with the help of effective and functional connectivity (Github)

4/25 - 5/25 **BCI & Neurotechnology Spring School**: organized by g.tec medical engineering GmbH. [Certificate]

AUG 2021 International **Interdisciplinary Computational Cognitive Science Summer School** .(Certificate)

06/2014 **Bharat Heavy Electricals Limited (BHEL)**, Plant -Durgapur project Ltd., WestBengal. (Certificate)

Report Advanced Automation and Instrumentation

06/15-07/15 Hands-on training on **Microcontroller based embedded system design** in Industrial Application by Oriens Infotech Pvt. Ltd. (Certificate)

Project Microcontroller based water level measurement system using tact-switches.

Technical Certifications

Python Programming (GUVI) [Certificate]; MATLAB, Image Processing & ODE (MathWorks) [Certificates]; C/C++ with Data Structures (CMC Academy) [Certificate]; Python for Data Science (Udemy) [Course] [Certificate]

Languages

English: IELTS Academic 7.0 (CEFR C1) —— Hindi, Bengali(Native) —— French(Elementary)

References

Prof. Jari Hyttinen
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